

Model review and revisions

JMP progress, September 17–30, 2026

Tommaso De Santo

I reviewed preferences, household finance, demographic accounting, and the empirical calibration inputs.

- Replaced the minimum-housing requirement with a first-child increase in the housing expenditure share.
- Allowed diminishing benefits from children at home, distinguishing them from children ever born.
- Separated children leaving home from adult household entry, and tied entrant assets to estates.
- Rebuilt the first-birth housing evidence and recalibrated the revised model.

Household decisions

Overlapping generations of households differ in age, earnings, wealth, housing tenure, and family size.

- During fertile ages, households choose whether to attempt a birth; success becomes less likely with age.
- They choose consumption, saving, housing size, and whether to rent or own; renters pay rent, while owners make a down payment and face selling costs.
- Persistent earnings risk and borrowing limits shape saving; a first-birth fixed cost and taste shocks shape fertility.
- Households retire, face mortality risk, and value bequests.

The interaction of housing costs and available wealth affects both birth timing and tenure choice.

Children and housing demand

Households value nonhousing consumption c , housing services s , and children currently at home m :

$$u(c, s, m) = \frac{[A(m)c^{\alpha(m)}s^{1-\alpha(m)}/e(m)]^{1-\sigma}}{1-\sigma} + v(m),$$

$$\alpha(m) = \alpha_0 - \Delta\alpha \mathbf{1}\{m > 0\}, \quad v(m) = \psi m^{1-\gamma}, \quad v(0) = 0.$$

Here m counts children at home and n is children ever born. $e(m)$ is the family-size equivalence scale, and σ is risk aversion.

$\alpha(m)$ is the consumption share; α_0 is the childless share and $\Delta\alpha$ is its reduction after the first child. ψ is the benefit level and γ its curvature.

$A(m)$ holds optimized material utility constant when the share changes, for given total expenditure and equivalence scale at the fixed reference rent. The first child raises the housing expenditure share; later children do not change that share.

Demographic and market equilibrium

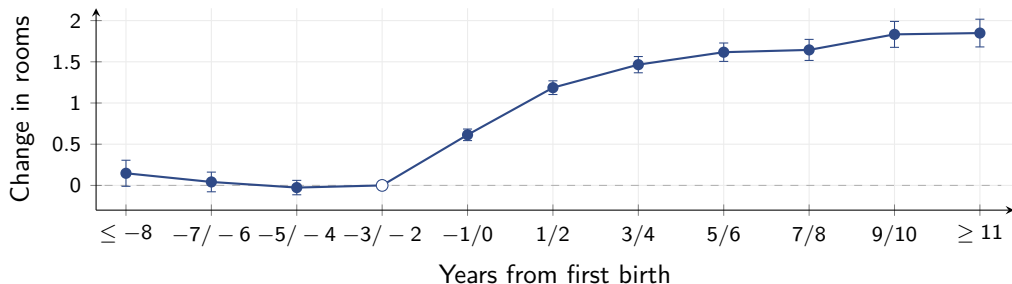
Children leaving home and new adult households entering the economy are separate transitions.

- Children leave dependency independently; adult entry follows earlier births with a lag.
- Positive estates finance fixed positive entrant assets; any remainder leaves the economy.
- The house price clears aggregate housing demand against housing supply.
- Equal retiree pensions are financed by a balanced payroll tax.

A stationary equilibrium has constant prices, allocations, and cohort structure, with demographic reproduction and market clearing.

Housing around the first birth

I rebuilt the PSID event study using interview dates and household status before the first birth.



At years 3–4, rooms rise by **1.465** (SE 0.050), ownership by **26.886 pp**, and recent moving falls by **21.147 pp**.

Reference person or spouse before birth. Baseline: years $-3/-2$; person and year effects; survey weights; 95% pointwise intervals.

Calibration strategy and inputs

I recalibrated the 2007 stationary economy after revising the empirical inputs.

Input	Value	Measurement
Mean occupied rooms	5.729	AHS 2007
Pension / working-household earnings	0.229	CPS ASEC, income 2007
Annual child-directed estates / wealth	0.729%	SCF 2007 mortality proxy
Annual depreciation / property tax	1.416% / 1.060%	National housing inputs

- The calibration matches fertility, wealth, housing size, and tenure; completed fertility is normalized to 2.1.
- The first-birth housing comparison still uses a model proxy for the empirical event study.

A period is four years. Each dependent child leaves with probability $2/9$; adult entry is split equally between 16 and 20 years after birth.

2007 stationary calibration

Moment	Target	Model
Completed fertility (normalization)	2.100	2.100
Childlessness, ages 40–44 (%)	19.828	20.106
Exactly one child among mothers, ages 40–44 (%)	21.366	20.942
Mean first-birth age (years)	25.976	25.933
First births at age 30+ (%) [†]	24.928	22.365
Wealth / earnings	6.927	6.326
Annual child-directed bequest flow / wealth (%)	0.729	0.706
Older wealth/income p90/p50 [†]	3.516	3.069
Mean occupied rooms	5.729	5.848
Ownership, ages 30–55 (%)	67.626	65.482
First-birth rooms difference	1.465	1.622
Family rooms difference [†]	0.385	0.353
Recent-parent ownership difference (pp)	12.761	11.955
Children per woman at age 25, capped at three	0.810	0.535

[†]Untargeted validation. Ten scored moments and a separate fertility normalization.

Calibrated parameters

The reference estimates ten free parameters and separately normalizes the child-benefit level.

Parameter	Economic interpretation	Value
H_0	Housing supply scale	6.294
β	Annual discount factor	0.963
χ	Owner housing-service premium	1.094
F_1	First-birth fixed utility cost	0.621
κ_1	First-birth taste dispersion	0.176
κ_+	Later-birth taste dispersion	0.332
θ_0	Bequest weight	0.125
$\Delta\alpha$	First-child housing-share loading	0.135
γ	Child-benefit curvature	0.061
κ_T	Tenure-choice dispersion	0.012
ψ	Child-benefit level (normalized)	0.136

Fixed inputs: $\sigma = 2$; $\alpha_0 = 0.733$; annual interest 2%; mortgage-financed share 80%.

Fertility at young ages

Subsequent experimental calibrations, with completed fertility normalized to 2.1:

Age-25 moment	Data	One birth per period	Two-birth option
Children per woman, capped at three	0.810	0.530	0.606
Mothers (%)	45.725	44.782	43.773
Children per mother, capped at three	1.770	1.185	1.385

- The share of mothers is close to the data; births among young mothers are too few.
- Allowing two births per period closes 27.1% of the child-count gap, with worse fit elsewhere.

The extension also adds a taste draw, so this comparison does not isolate birth spacing.

Credit and further calibration

I reviewed the borrowing rules and tested alternative initial-wealth distributions at a common 2% annual real interest rate.

Entrant wealth rule	Renter debt limit (mean annual earnings)
Empirical wealth/earnings ratios drawn independently of earnings	0.25
Zero entrant wealth	0
Nonnegative ratio distribution with the original mean	0

- The three pilot calibrations are complete; an entrant-wealth and credit specification has yet to be selected.
- The next step is to resolve that choice and assess the remaining young-fertility gap.

The stationary solver was reorganized and accelerated to support calibration.